of stochastic perturbation). Unlike for r-RESPA, the splitting needs to be performed at the level of the differential equations. We suppress the index of the degree of freedom, takes its mass to be one for simplicity of presentation, and write the equations of motion for the variables associated to a single physical degree of freedom in the form

$$\begin{aligned}
\mathrm{d}q &= p\mathrm{d}t,\\
\mathrm{d}p &= F\mathrm{d}t - \left(\frac{p(F^{\mathrm{F}}+F^{\mathrm{S}}) - \frac{1}{2}\mu_1\xi_1^2\xi_2}{\Lambda}\right)p\mathrm{d}t,\\
\mathrm{d}\xi_1 &= -\left(\frac{p(F^{\mathrm{F}}+F^{\mathrm{S}}) - \frac{1}{2}\mu_1\xi_1^2\xi_2}{\Lambda}\right)\xi_1\mathrm{d}t - \xi_2\xi_1\mathrm{d}t,\\
\mathrm{d}\xi_2 &= \mu_2^{-1}(\mu_1\xi_1^2 - \beta^{-1})\mathrm{d}t - \gamma\xi_2\mathrm{d}t + \sigma\mathrm{d}W,
\end{aligned}$$

where $\Lambda = p^2 + \frac{1}{2}\mu_1\xi_1^2$. $\Lambda = \beta^{-1}$ is constant of motion.

Because of the form of the coupling, we observe that this vector field can be additively decomposed based on the various terms in the stochastic isokinetic equations, including the fast-slow splitting of the forces, into the following generators:

Kinetic part:

$$\mathcal{L}_{\mathrm{K}} = p\frac{\partial}{\partial q},$$

isokinetic Fast force:

$$\mathcal{L}_{\mathrm{F}} = \left[F^{\mathrm{F}} - (p^2/\Lambda)F^{\mathrm{F}}\right]\frac{\partial}{\partial p} - \frac{p\xi_1 F^{\mathrm{F}}}{\Lambda}\frac{\partial}{\partial \xi_1},$$

isokinetic Slow force:

$$\mathcal{L}_{\mathrm{S}} = \left[F^{\mathrm{S}} - (p^2/\Lambda)F^{\mathrm{S}}\right]\frac{\partial}{\partial p} - \frac{p\xi_1 F^{\mathrm{S}}}{\Lambda}\frac{\partial}{\partial \xi_1},$$

isokinetic Nosé terms:

$$\mathcal{L}_{\mathrm{N}} = \frac{1}{2}\frac{\mu_1\xi_1^2\xi_2 p}{\Lambda}\frac{\partial}{\partial p} + \frac{1}{2}\frac{\mu_1\xi_1^3\xi_2}{\Lambda}\frac{\partial}{\partial \xi_1} - \xi_2\xi_1\frac{\partial}{\partial \xi_1} + \mu_2^{-1}[\mu_1\xi_1^2 - \beta^{-1}]\frac{\partial}{\partial \xi_2},$$

Ornstein-Uhlenbeck term:

$$\mathcal{L}_{\mathrm{O}} = -\gamma\xi_2\frac{\partial}{\partial \xi_2} + \frac{\sigma^2}{2}\frac{\partial^2}{\partial \xi_2^2}.$$

It is straightforward to show that each of the terms of the corresponding vector field splitting preserves the first integral $\Lambda =$ constant. For example

$$\begin{aligned}\mathcal{L}_{\mathrm{F}}\Lambda &= \left[F^{\mathrm{F}} - (p^2/\Lambda)F^{\mathrm{F}}\right]\frac{\partial \Lambda}{\partial p} - \frac{p\xi_1 F^{\mathrm{F}}}{\Lambda}\frac{\partial \Lambda}{\partial \xi_1} \\ &= \left[F^{\mathrm{F}} - (p^2/\Lambda)F^{\mathrm{F}}\right](2p) - \frac{p\xi_1 F^{\mathrm{F}}}{\Lambda}(\mu_1\xi_1) \\ &= pF^{\mathrm{F}}\left[2 - \Lambda^{-1}(2p^2 + \mu_1\xi_1^2)\right] \\ &= 2pF^{\mathrm{F}}\left[1 - \Lambda^{-1}\Lambda\right] = 0.\end{aligned}$$

The proof for the other terms is similar.

The given decomposition can be used to construct a variety of algorithms, for example the following which we term the Stochastic Isokinetic Nosé-Hoover RESPA algorithm.

SIN(R) Algorithm

Propagate p, ξ_1, ξ_2 by $\exp\left[(h/2)\mathcal{L}_{\mathrm{N}}\right]$.
Propagate p, ξ_1 by $\exp\left[(\tilde{h}/2)\mathcal{L}_{\mathrm{F}} + (h/2)\mathcal{L}_{\mathrm{S}}\right]$, $\quad (\tilde{h} = h/r)$.
Propagate q by $\exp\left[\tilde{h}\mathcal{L}_{\mathrm{K}}\right]$.
Propagate ξ_2 by $\exp\left[\tilde{h}\mathcal{L}_{\mathrm{O}}\right]$.
Propagate p, ξ_1 by $\exp\left[(\tilde{h}/2)\mathcal{L}_{\mathrm{F}}\right]$.
Repeat $r - 2$ times:
  Propagate p, ξ_1 by $\exp\left[(\tilde{h}/2)\mathcal{L}_{\mathrm{F}}\right]$.
  Propagate q by $\exp\left[(\tilde{h}/2)\mathcal{L}_{\mathrm{K}}\right]$.
  Propagate ξ_2 by $\exp\left[(\tilde{h})\mathcal{L}_{\mathrm{O}}\right]$.
  Propagate q by $\exp\left[(\tilde{h}/2)\mathcal{L}_{\mathrm{K}}\right]$.
  Propagate p, ξ_1 by $\exp\left[(\tilde{h}/2)\mathcal{L}_{\mathrm{F}}\right]$.
Propagate p, ξ_1 by $\exp\left[(\tilde{h}/2)\mathcal{L}_{\mathrm{F}}\right]$.
Propagate ξ_2 by $\exp\left[\tilde{h}\mathcal{L}_{\mathrm{O}}\right]$.
Propagate q by $\exp\left[\tilde{h}\mathcal{L}_{\mathrm{K}}\right]$.
Propagate p, ξ_1 by $\exp\left[(h/2)\mathcal{L}_{\mathrm{F}} + (h/2)\mathcal{L}_{\mathrm{S}}\right]$.
Propagate p, ξ_1, ξ_2 by $\exp\left[(h/2)\mathcal{L}_{\mathrm{N}}\right]$.